NAD+

Nicotinamide Adenine Dinucleotide | 500mg & 1,000mg x 10 Vials

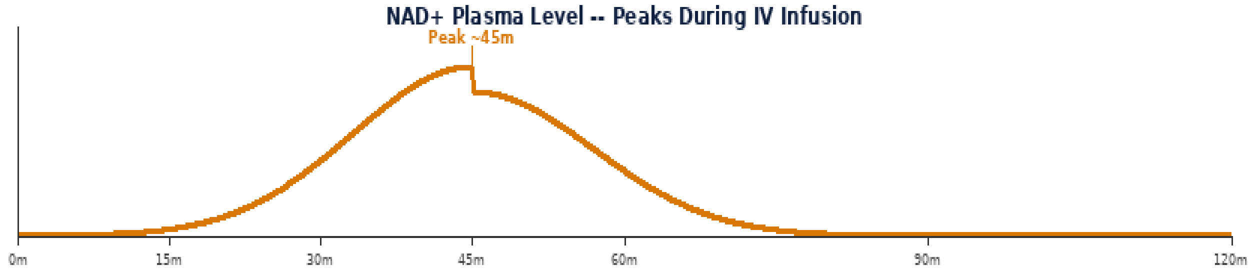
Cellular Energy | Sirtuin Activation | IV / IM / SubQ | Use within 24hr

Purity: 99.8% | Endotoxin-Free | Research Use Only

NAD+ (Nicotinamide Adenine Dinucleotide) is a coenzyme essential in virtually every metabolic pathway. Intracellular NAD+ declines ~50% between ages 20-60. Supplementation activates sirtuins (SIRT1-7), PARP DNA repair, and restores mitochondrial function. IV delivery achieves supraphysiological plasma levels impossible with oral NMN or NR precursors.

Parameter	Specification
Class	Coenzyme -- NAD+/NADH redox pair, sirtuin cofactor
Vial Sizes	500mg x 10 | 1,000mg x 10 | Purity 99.8%
Reconstitution	500mg: 10mL saline or BAC->50mg/mL | 1,000mg: 10mL->100mg/mL
Standard Dose	500-1,000mg IV drip (30-60 min) | IM or SubQ lower doses
Route	IV drip (primary) | IM | SubQ | NEVER rapid IV bolus
CRITICAL	Use within 24 hours of reconstitution -- NAD+ degrades rapidly in solution
Frequency	IV: 1-3x per week | IM/SubQ: daily if needed | no fasting required

MECHANISM OF ACTION & RESEARCH BENEFITS

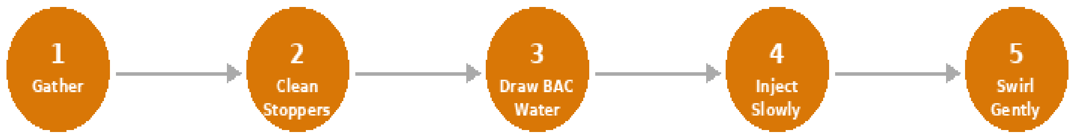


Pathway	Mechanism	Benefit
Sirtuin Activation	NAD+ is obligate cofactor for SIRT1-7 (class III HDACs); SIRT1 deacetylates PGC-1alpha, p53, NF-kB; regulates mitochondrial biogenesis, inflammation, DNA repair, and metabolic sensing	Anti-aging; metabolic improvement; mitochondrial function; gene regulation
PARP DNA Repair	PARP1/2 consume NAD+ for poly-ADP-ribosylation; essential for single-strand DNA break repair; high DNA damage (aging, UV, oxidative stress) depletes NAD+; restoring NAD+ enhances repair capacity	DNA repair; genomic stability; cancer prevention research
Mitochondrial Function	NAD+/NADH ratio drives electron transport chain efficiency; restoring NAD+ improves complex I activity; enhances ATP production; reduces oxidative stress from mitochondrial dysfunction	Energy restoration; fatigue; mitochondrial disease research

Benefit	Context
Sirtuin Activation	NAD+ is the mandatory cofactor for all 7 sirtuins -- the most studied longevity enzymes. IV NAD+ achieves activation impossible with oral precursors.
Anti-Aging Research	Declining NAD+ is one of the most consistent hallmarks of aging. IV supplementation reverses cellular markers of aging in research models.
Energy & Cognitive	Rapid restoration of energy metabolism and cognitive function reported in IV NAD+ research. Effects typically felt during the infusion.

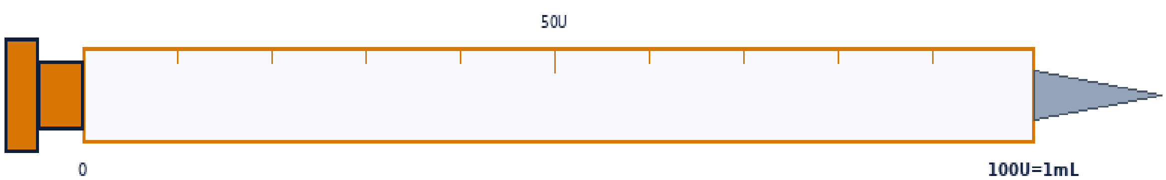
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



1mL Insulin Syringe | 29-31G | SubQ

Formula: Vol(mL)=Dose(mcg)/Conc(mcg/mL) | **Units:** Vol x 100

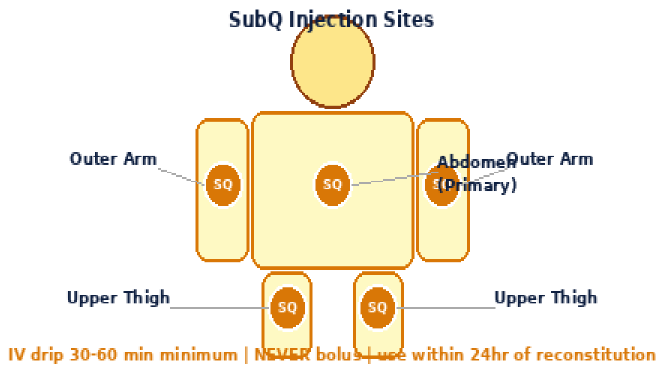
500mg x 10 -- 10mL BAC -> 50.0mg/mL (50,000mcg/mL)

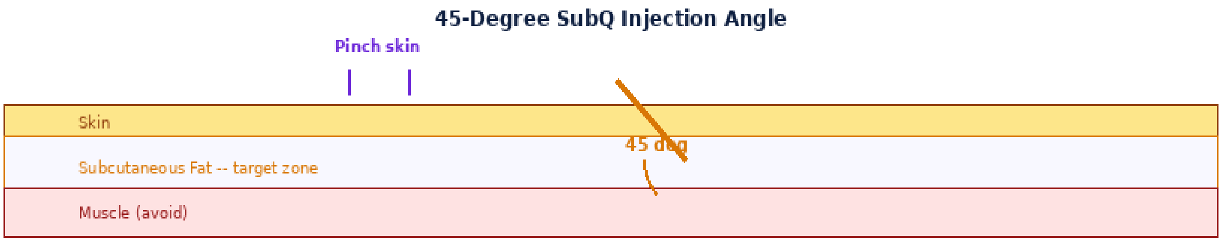
Dose(mcg)	500mcg	1000mcg	2000mcg	5000mcg
Vol(mL)	0.0100	0.0200	0.0400	0.1000
Units	1.0U	2.0U	4.0U	10.0U
Doses/Vial	100	50	25	10

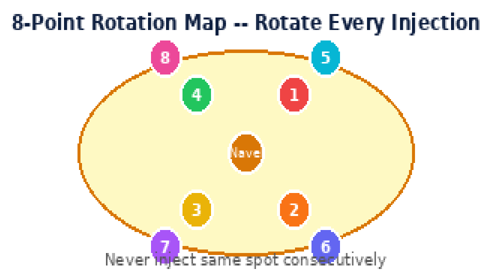
1000mg x 10 -- 10mL BAC -> 100.0mg/mL (100,000mcg/mL)

Dose(mcg)	500mcg	1000mcg	2000mcg	5000mcg
Vol(mL)	0.0050	0.0100	0.0200	0.0500
Units	0.5U	1.0U	2.0U	5.0U
Doses/Vial	200	100	50	20

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES







Step	Action	Detail
1	Reconstitute	Add 10mL saline to vial. Swirl gently. Must be clear.
2	IV drip	Dilute dose in 250-500mL 0.9% NaCl. Infuse over 30-60 min minimum. SLOW infusion prevents side effects.
3	IM/SubQ option	Lower doses (100-500mg): IM deltoid or SubQ abdomen. 23-25G IM, 29-31G SubQ.
4	Use within 24hr	Reconstituted NAD+ degrades rapidly. Prepare fresh each session. Discard unused.

Protocol	Dose	Frequency	Duration	Notes
IV Therapy	500-1,000mg	1-3x/week (IV drip)	Ongoing w/ monitoring	Infuse over 30-60 min minimum.
IM/SubQ	100-500mg	Daily or 3-5x/week	Ongoing	Lower dose. More practical for self-admin.

Side Effect	Likelihood	Management
Nausea / flushing (IV)	Very Common	RATE-related. Slow infusion to 60 min. Resolves completely.
Chest tightness (IV)	Common (fast)	Slow infusion immediately. Never bolus. Resolves with rate reduction.
Muscle cramps (IV)	Common	Add magnesium to IV bag. Slow rate.
Fatigue post-IV	Occasional	Post-infusion tiredness is normal. Rest after session.

Contraindications: Cancer history, pregnancy, under 18.

1. 500mg vial: 10mL diluent->50mg/mL | 1,000mg vial: 10mL->100mg/mL.
2. CRITICAL: Use reconstituted NAD+ within 24 hours -- degrades rapidly in solution.
3. IV drip: dilute in 250-500mL saline. Infuse over MINIMUM 30-60 minutes.
4. NEVER administer NAD+ as a rapid IV bolus -- causes flushing, nausea, chest tightness.
5. All IV side effects are rate-related: slow the infusion and they resolve.
6. IM and SubQ routes tolerated at lower doses (100-500mg) without major side effects.
7. No fasting required. Inject or infuse any time of day.
8. Store unreconstituted at 2-8 deg C. Prepare fresh each session -- do not refrigerate reconstituted overnight.
9. Contraindicated in known NAD+ sensitivity. Monitor liver function on long-term protocols.